

MAE 2250: Introduction to Mechanical Design

Open Design Project 5: Prototype Design and Testing Report

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1 Overview

This report documents the design, assembly, and testing of our functional prototype for mechanically removing spotted lanternfly (SLF) models from grape clusters. The prototype includes a rotating brush mounted on a shaft powered by a drill and a manually simulated conveyor system.

2 Parts List and Specifications

2.1 Shaft

- McMaster Code: 8920K231
- Unit Cost: \$9.77
- Length: 1 ft
- Length Tolerance: -0.005" to 0"
- Diameter 1"
- Diameter Tolerance: -0.002" to 0"
- Material: Low-Carbon Steel, Grade 1018
- Shape: Rod and Disc
- Yield Strength: 54,000 psi
- Fabrication: Cold Worked
- Hardness: Rockwell B70
- Heat Treatable: Yes
- Certificate: Material Certificate with Traceable Lot Number
- Specifications Met: ASTM A108
- Straightness Tolerance: Not Rated
- Coefficient of Thermal Expansion: 7.1×10^{-6}
- Elongation: 15%
- Material Composition: C 0.13-0.20%, Mn 0.30-0.90%, P 0.04% max, Si 0.15-0.30%, S 0.50% max, Fe remainder
- Additional Specifications: SDS, RoHS, REACH, DFARS
- Country of Origin: USA
- Fabrication Notes: Machined 1.75 in from one end to 0.48 in diameter to fit inside drill chuck.

2.2 Brush

- McMaster Code: 7442T11
- Unit Cost: \$8.95
- Brush Length: 1ft
- Bristle Material: Polyester Plastic
- Backing Material: Stainless Steel
- Bristle Dimensions: 3/16" W $\times$ 7/32" H, Overall Diameter: 1", Bristle Diameter: 0.008"
- Color: White
- Max Temperature: 200°F
- Compliance: FDA 21 CFR 177.1660

2.3 Drill

- Model: Dewalt Drill
- Source: Taylor Design Studio

2.4 Conveyor Simulation Components

- Cardboard Piece
 - Simulates conveyor surface
 - Source: Taylor Design Studio
- Wooden Boards
 - Support frame
 - Source: Taylor Design Studio
- Zip Ties
 - Secure components
 - Source: Taylor Design Studio
- Glue
 - Temporary bonding
 - Source: Taylor Design Studio

2.5 Assembly Sketches and Photos

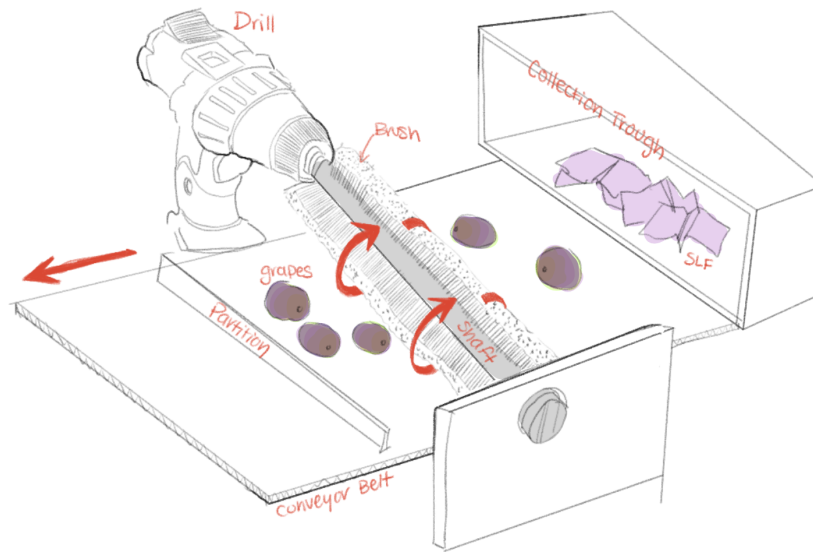


Figure 1: Annotated sketch of prototype. Arrows indicate brush rotation and conveyor motion.

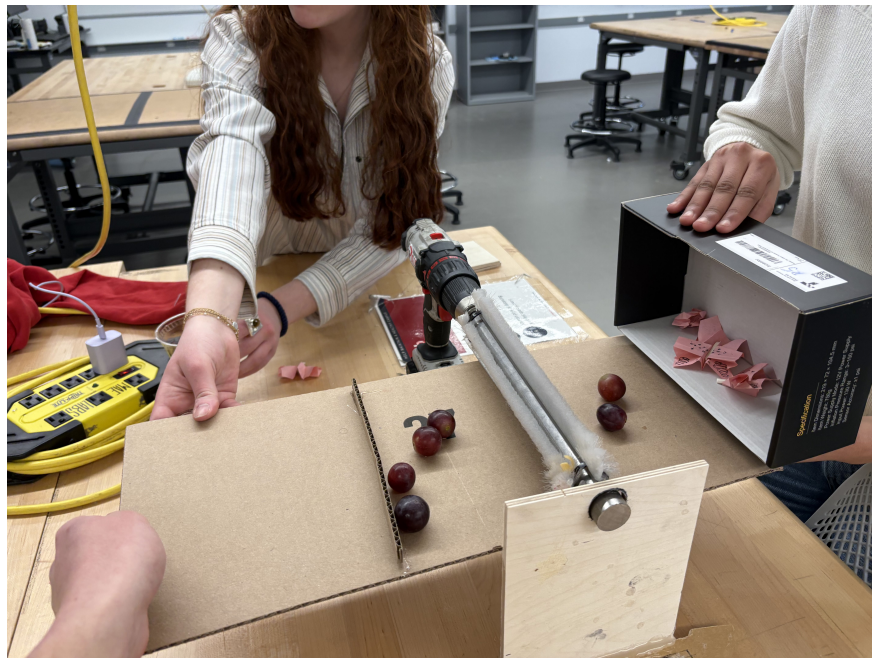


Figure 2: Photo of assembled prototype showing brush, shaft, and conveyor simulation.

3 Assembly Instructions

3.1 Step 1: Shaft Preparation

1. Cut 1 ft low-carbon steel rod.
2. Machine 1.75 in from one end to 0.48 in diameter using lathe (RPL) to fit the drill chuck.
3. Deburr and clean the shaft.

3.2 Step 2: Brush Mounting

1. Cut the stock strip brush into three equal-length sections.
2. Position the sections so they are evenly spaced along the shaft for uniform bristle coverage.
3. Secure each brush section with zip ties wrapped tightly around the shaft and brush backing.
4. Apply a thin layer of glue between the brush backing and the shaft to prevent axial movement.
5. Inspect alignment to ensure bristles are perpendicular to shaft axis.

3.3 Step 3: Drill Interface

1. Insert machined shaft into drill chuck.
2. Tighten chuck to prevent slippage during testing.

3.4 Step 4: Shaft Mount

1. Prepare a wooden mounting board by drilling a 1-inch diameter hole to accommodate the shaft.
2. Position the hole so that its center is offset 1-inch from the top edge of the board.
3. Secure this mounting board to a thicker base board using wood glue, ensuring it is perfectly perpendicular to the base for proper shaft alignment.
4. Allow the glue to fully cure before proceeding with mounting the shaft.

3.5 Step 5: Conveyor Simulation Setup

1. Place cardboard piece underneath the base of the brush.
2. Install a partition on the cardboard to separate grapes from dislodged SLF.
3. Ensure smooth sliding surface for grape clusters.

3.6 Step 6: Collection Trough

1. Place a cardboard box behind the brush to collect SLF.

3.7 Step 7: System Integration

1. Place grape clusters on conveyor surface.
2. Operate drill to rotate brush at target RPM (450–500 RPM).
3. Slide the cardboard manually to simulate conveyor motion.

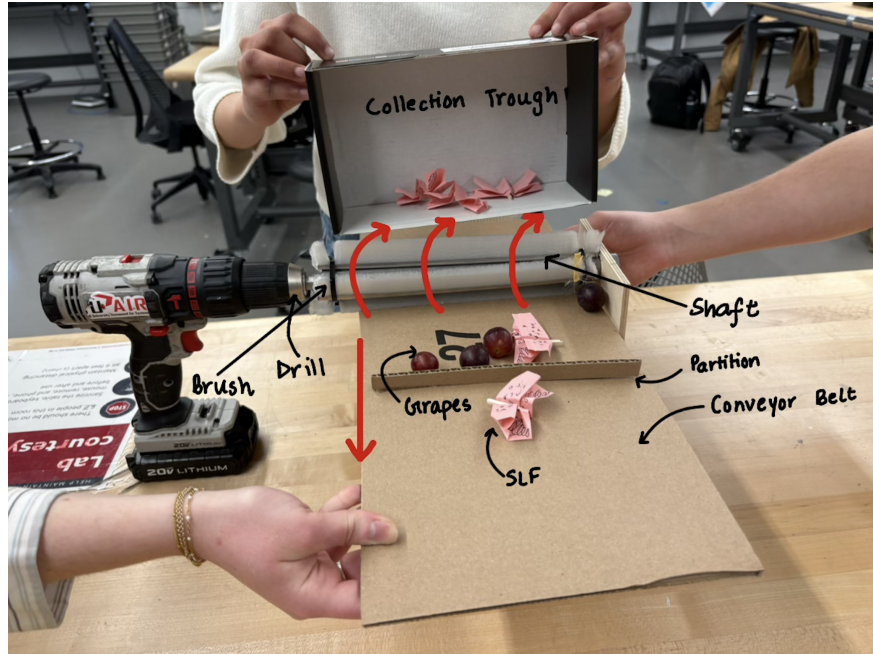


Figure 3: Visual representation of functional connections.

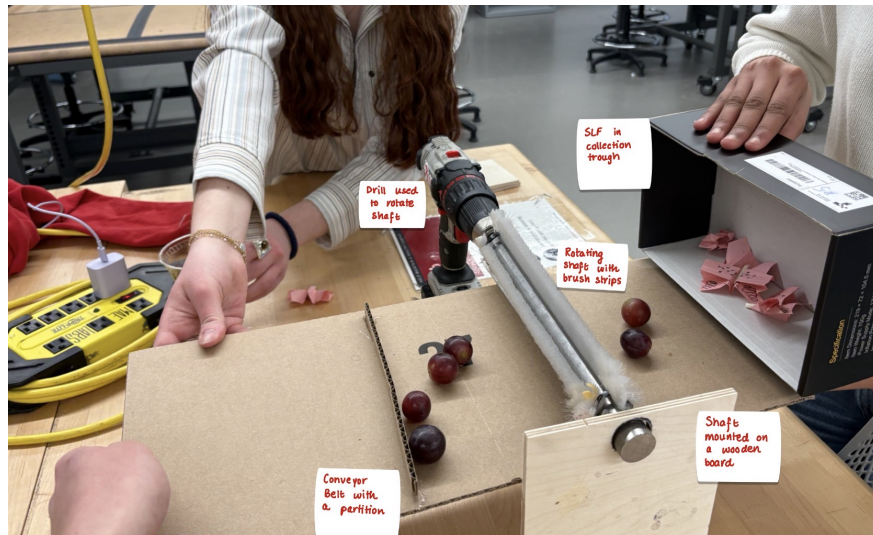


Figure 4: Annotated parts of the prototype showing brush, shaft, and conveyor components.

4 Design Tests

4.1 Test 1: Brush Rotation Speed

Part Tested: Shaft-mounted brush assembly

Purpose: Determine optimal brush speed to remove SLF models without damaging grape clusters.

Method: The drill was operated at 450, 500, 550, 600, and 650 RPM. At each speed, a cluster of 20 grape models with attached SLF paper models was passed under the rotating brush. The number of SLF models removed and any grape displacement or damage was recorded.

Results:

- 450 RPM: 6/7 SLF removed, negligible grape displacement.
- 500 RPM: 3/7 SLF removed, minimal grape movement.
- 550 RPM: 3/7 SLF removed, slight grape displacement.
- 600 RPM: 1/7 SLF removed, moderate grape displacement.
- 650 RPM: 1/7 SLF removed, significant grape displacement; clusters moved off cardboard.

Conclusion: Target rotation speed for optimal SLF removal with minimal grape displacement is 500 RPM. Future iteration will include an adjustable speed drill or motor with RPM control.

4.2 Test 2: Brush Bristle Coverage

Part Tested: Three-section strip brush mounted on shaft

Purpose: Evaluate uniformity of bristle coverage along the shaft to ensure consistent SLF removal.

Method: Paper SLF models were placed along the entire width of grape clusters. Brush rotated at 500 RPM, and the number of SLF models removed from each section (left, center, right) was recorded.

Results:

- Left section: 5/7 removed
- Center section: 7/7 removed
- Right section: 4/7 removed

Minor gaps at ends due to spacing between brush sections.

Conclusion: Adjust brush spacing and overlap sections slightly in future iteration to achieve ~90% SLF removal across full width.

4.3 Test 3: Conveyor Simulation Stability

Part Tested: Cardboard conveyor simulation

Purpose: Ensure stable grape movement and proper SLF collection.

Method: Cardboard manually slid beneath the rotating brush with 10 grape clusters. Observed for tipping, sliding, or grapes falling off.

Results:

- 6/10 clusters remained fully on conveyor during all passes.
- 4/10 clusters experienced grapes falling off due to slight flexing and lack of lateral support.

- SLF models removed as expected, but grape displacement occurred where clusters fell off.

Conclusion: The current conveyor design allows some grape loss. In the next prototype, adding side walls or guides along the conveyor will prevent grapes from falling off and improve stability. Reinforcing the cardboard with a thicker backing or add light wooden boards on either side would ensure stability.

4.4 Test 4: SLF Removal Efficiency

Part Tested: Integrated system: brush, shaft, manual conveyor

Purpose: Quantify overall SLF removal efficiency for a single pass.

Method: 20 grape clusters, each with 20 SLF paper models, passed under brush at 500 RPM while sliding cardboard at 1 in/sec. Counted SLF removed per cluster.

Results:

- Average SLF removal: 72%
- Range: 65%–78% per cluster
- No visible grape damage noted.

Conclusion: System is moderately effective. Increasing bristle density, overlapping brush sections, and introducing consistent conveyor motion will improve removal to 90%.

4.5 Test 5: Assembly Repeatability and Stability

Part Tested: Shaft mounting, brush sections, drill interface, conveyor base

Purpose: Evaluate ease of assembly, alignment, and structural stability for repeated use.

Method: Prototype was assembled and disassembled 5 times by two different team members. Time to assemble, fit of shaft in drill, brush security, and cardboard stability were recorded.

Results:

- Average assembly time: 9 minutes
- Shaft fit consistently in drill chuck with no slippage
- Brush sections remained secure; minor shifting of zip ties observed
- Cardboard conveyor stable, minor flexing noted

Conclusion: Adding alignment markings on shaft and brush sections and using tighter zip ties or small clamps will improve repeatability. Prototype is sufficiently stable for testing, but next iteration should reinforce conveyor base.

5 Success Criteria

The prototype must efficiently remove SLF models while remaining safe and repeatable for testing.

- **SLF Removal Efficiency:** At least 90% removal of SLF models after one pass. Measured by counting removed models. High priority.
- **Grape Integrity:** No visible damage to grapes; assessed visually. Medium priority.

- **Assembly Time:** Prototype can be assembled in under 10 minutes by one user following instructions. Measured using stopwatch. Medium priority.
- **Conveyor Consistency:** Grapes move smoothly along simulated conveyor at a consistent speed of 0.5–1 in/sec. Measured using ruler and stopwatch. Demonstrable on exhibit day. High priority.
- **Brush Speed Control:** Brush rotates within 450–500 RPM range; measured with tachometer. High priority.